# Willow Creek Bridge Renewal

## Deck drainage geometry and outstanding approvals

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|---|---|
| CLIENT / OWNER | Prairie County Transportation (fictional) |
| DOCUMENT / REVISION | WC-TN-003 / P01 - preliminary |
| LOCATION / STATUS | Rural Alberta - fictional site / demonstration concept |

### Geometry checks

At 1.0% crossfall, the fall from crown to either carriageway edge over a 4.0 m half-width is 40 mm. At 2.0%, the corresponding fall is 80 mm. These geometric calculations do not replace a drainage design.

### Profile check

A 0.35% constant grade over 24 m gives an 84 mm longitudinal level difference. The concept does not include a crest curve or super-elevation exception.

### Approval record

No approval for a longitudinal grade below 0.5% has been obtained or included in this package. The approach constraint is an owner preference, not an approved departure from the adopted bridge criteria.

### Outstanding investigations

A site-specific hydraulic assessment, flood levels, scour assessment and geotechnical investigation are outstanding. No freeboard, foundation capacity or final hydraulic adequacy claim is made.